# A Practical Guide to Writing in Physics Courses

Ray D. Sameshima

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## 0 Start Here

This guide is meant to be used throughout the course rather than read once and forgotten. Before submitting written work, check the following basic expectations.

- ☐ **Identify yourself and the course clearly.** In professional email, include your full name and course information.
- ☐ **Show your reasoning.** A numerical answer alone is usually not a complete physics solution. Define quantities, state the relevant relations, and make the main steps of your reasoning visible.
- ☐ **Define symbols and use units consistently.** Every symbol should be introduced before use, and numerical results should carry appropriate units and significant figures.
- ☐ **Make your work reproducible.** A reader should be able to follow what you did, how you obtained the result, and why the conclusion follows.
- ☐ **For laboratory reports, write for a reader who was not present.** Ask: *Can a classmate who missed the lab understand the physics, procedure, data analysis, and conclusion from my report alone?*
- ☐ **Use sources and outside tools responsibly.** Borrowed words, ideas, figures, or data must be acknowledged as appropriate. Follow the course syllabus for collaboration and generative-AI rules.

The later sections provide more detailed checklists and examples for email, problem solutions, mathematical arguments, and laboratory reports.

## 1 Introduction

Effective scientific writing is a core skill in academic settings. Whether communicating with professors, presenting homework solutions, or preparing lab reports, clear writing reflects clear thinking. In science courses, the focus must be on concise explanations, explicit reasoning, correct notation, and reproducible methods. This guide provides practical tips to help you meet these expectations with clarity and precision.

The following list includes commonly used writing tools in physics, especially those that support mathematical and equation-editing capabilities:

- LaTeX[1]
- Google Docs[2]: See [Goo] for further instructions.
- Microsoft Word[3]: See [Mic] for further instructions.

[1] `https://www.latex-project.org`
[2] `https://docs.google.com`
[3] `https://www.microsoft.com/en-us/microsoft-365/word`

## 2 E-Mail Writing

Professional communication via email is essential in academic settings, especially when contacting professors, TAs, or other academic staff.

- ☐ You identify yourself.
  - ☐ You are using your professional, institutional, or **school email**.
  - ☐ Your full name appears in the signature block.
  - ☐ Your email includes your course identifier: course id and date&time.
- ☐ Subject Line includes **course id**.
  - ☐ The subject line briefly describes your request/question.
- ☐ Body is well structured.
  - ☐ The opening contains the **recipient's name** with an appropriate greeting.
  - ☐ The next line briefly introduces who you are: **full name and course details**.
  - ☐ You focus on a single main topic.
  - ☐ You state your request or question explicitly and politely.
  - ☐ You end with an appropriate sign-off with your full name.
- ☐ Attachment, if any
  - ☐ You mention each attachment in the body.
  - ☐ All attachments are correctly named and readable.
- ☐ You have proofread the address(es) and your email.

Allow at least 24-48 hours for replies, especially during busy periods. For further examples of best practices in professional email communication, see [Rob20]. An example of poor email etiquette is illustrated in Figure 1.

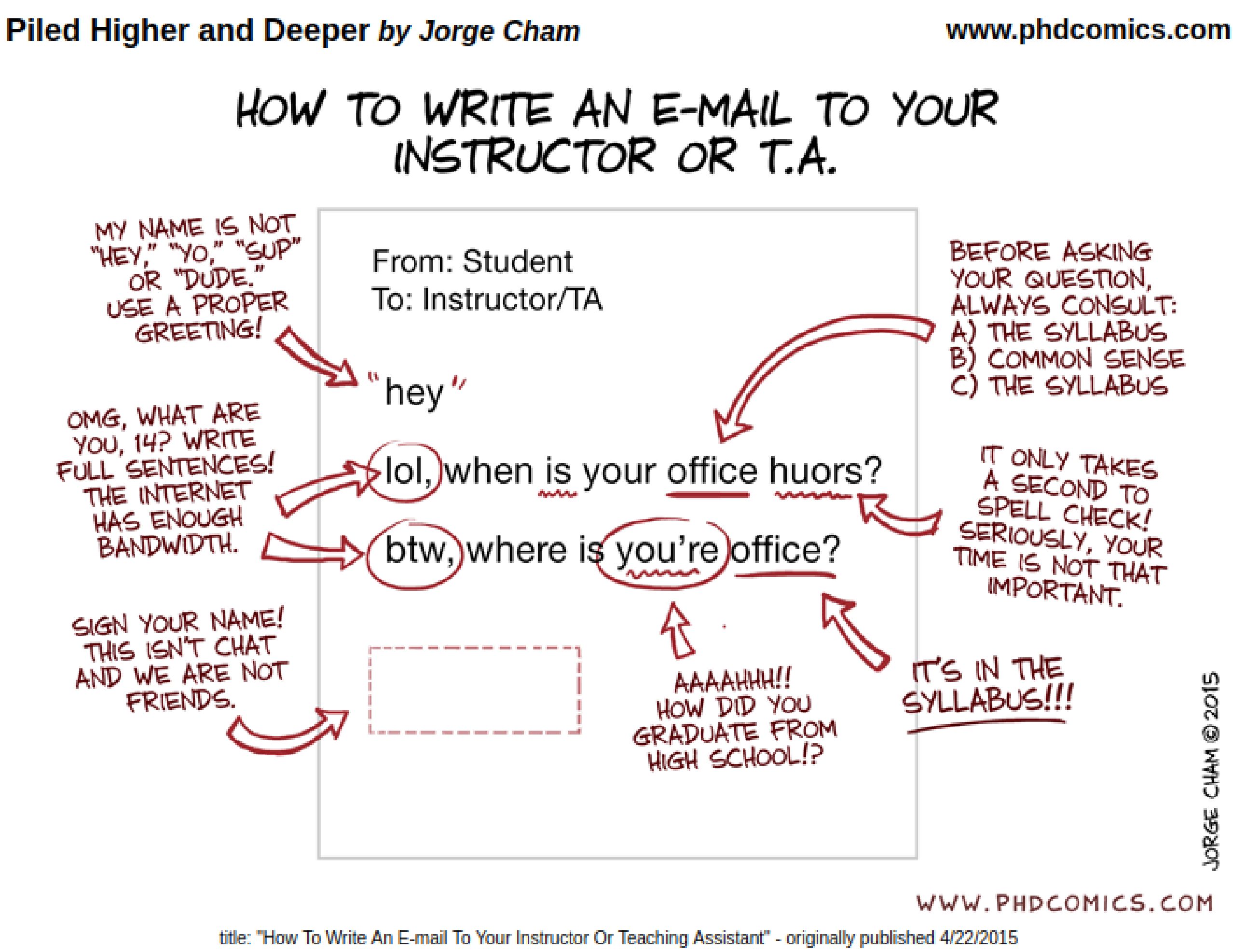


Figure 1: Example of an unacceptable email, adapted from [Cha15].

# 3 Solution Writing

A good solution guides your readers through your reasoning without forcing them to fill in large gaps.

## 3.1 Solve, Substitute, and Round-off

When tackling problem-based exams, first make sure you understand what the problem is asking. Then plan your approach, carry it out, and check your work.

**Example 1.** Let us look at a free-fall motion starting from rest. The position of the falling object at time $t$ is given by $-\frac{|g|}{2}t^2$ and the velocity is $-|g|\,t$, where $|g| = 9.81\,\mathrm{m\,s^{-2}}$ is the magnitude of downward gravitational acceleration. Find the falling object's velocity at height $h = -10.0\,\mathrm{m}$.

1. Identify knowns and relevant equations:
   The object's position at time $t$, relative to the initial point, is $-\frac{|g|}{2}t^2$. We know $|h| = 10.0\,\mathrm{m}$ and $|g| = 9.81\,\mathrm{m\,s^{-2}}$.
2. Solve:
   Let $T$ be the time when the object reaches $h = -10.0\,\mathrm{m}$:

$$-\frac{|g|}{2}T^2 = -|h|\,. \tag{1}$$

   Solve for the time of fall $T$. Then we get $T = \sqrt{\frac{2\,|h|}{|g|}}$. The corresponding velocity is $-|g|\,T = -|g|\sqrt{\frac{2\,|h|}{|g|}} = -\sqrt{2\,|h|\,|g|}$.
3. Substitute and round off:
   The speed at $h = -10.0\,\mathrm{m}$ is:

$$\begin{aligned}\sqrt{2\,|h|\,|g|} &= \sqrt{2 \times 10.0\,\mathrm{m} \times 9.81\,\mathrm{m\,s^{-2}}}\\ &= \sqrt{2 \times 10.0 \times 9.81}\mathrm{m\,s^{-1}}\\ &= 14.007\cdots\mathrm{m\,s^{-1}}\\ &\approx 1.40 \times 10^1\,\mathrm{m\,s^{-1}}.\end{aligned} \tag{2}$$

   Therefore, the velocity at height $h = -10.0\,\mathrm{m}$ is $\boxed{-1.40 \times 10^1\,\mathrm{m\,s^{-1}}}$.

Many textbooks, for instance [UH22], contain fully solved examples, see Example 2.14 in OpenStax College Physics 2e §2.7 Falling Objects.

**Quick Self-Check**

- ☐ All symbols and abbreviations and givens are defined before use.
- ☐ Units are consistent. (In SI: m, kg, s, A, etc.)
- ☐ Final answers use appropriate significant figures and are visually highlighted, using underline, **boldface**, or $\boxed{\text{box}}$, for example.
- ☐ The signs (±) and order of magnitude are checked.

## 3.2 Mathematical Proofs

Understanding mathematical proofs can be challenging, and writing them can be even more so. Some problems require students to demonstrate mathematical reasoning through a formal proof.

### 3.2.1 Elements of Mathematical Proofs

A mathematical proof must be self-contained and clearly present the logical argument.

According to [Cen21], many mathematical propositions are universally quantified implications of the form "for all (object of a certain type), if (hypothesis), then (conclusion)." Symbolically, it is given by

$$\forall x : H(x) \Rightarrow C(x), \tag{3}$$

where $x$ is the universally quantified variable, $H(x)$ is the hypothesis, and $C(x)$ is the conclusion. Proofs for this type of statement must begin with "Let $x$ be ...."

Here are three common proof strategies:

1. Direct Proof – Often the preferred method whenever possible:
   Assume the hypothesis $H(x)$ is true and, using this assumption, definitions, and previously proven results, logically deduce the conclusion $C(x)$.
2. Proof by Contrapositive:
   Recall that $p \Rightarrow q$ – if $p$ then $q$ – is logically equivalent to $\neg q \Rightarrow \neg p$, see Table 1.

| $p$ | $q$ | $p \Rightarrow q$ |
|---|---|---|
| True | True | True |
| True | False | False |
| False | True | True |
| False | False | True |

Table 1: The truth table for the implication $p \Rightarrow q$. Since both $p \Rightarrow q$ and $\neg p \vee q$ – not $p$ or $q$ – are false precisely when $p$ is true and $q$ is false, $p \Rightarrow q$ is logically equivalent to $\neg p \vee q$.

   Note that $\neg$ is the negation, also called logical not, $\vee$ is the logical or, and $\wedge$ is the logical and. Therefore, to prove $H(x) \Rightarrow C(x)$ for $x$, we can instead assume $\neg C(x)$ and prove $\neg H(x)$.
3. Proof by Contradiction:
   Assume the implication is false, i.e., $\neg (H(x) \Rightarrow C(x))$, which is equivalent to $H(x) \wedge \neg C(x)$. Use both assumptions $H(x)$ and $\neg C(x)$ to derive a contradiction.

   > ...in other words, you are indirectly proving that the original implication is true by showing that it cannot possibly be false.[Cen21]

**Remark.** *To disprove a universally quantified statement:*

$$\forall x : P(x), \tag{4}$$

*where $P(x)$ is a proposition, we should find a counterexample – a specific $x_0$ for which $P(x_0)$ is false. A single clear counterexample is sufficient to disprove a universally quantified statement.*

**Example 2.** We showcase examples of three strategies:

1. For each $n \in \mathbb{N}$, $n(n+1)$ is even.
   Let $n \in \mathbb{N}$. Then, $n$ is either odd or even; if $n$ is odd, then $n+1$ is even, otherwise $n$ is even. Hence, the product $n(n+1)$ is even.
2. For $m, n \in \mathbb{N}$, we say they have the same parity if either both are even or both are odd. If $m+n$ is even, then $m$ and $n$ have the same parity.
   The contraposition of the statement is "if $m$ and $n$ are different parity, then $m+n$ is odd." Suppose $m$ and $n$ are of different parity; without loss of generality, we may assume $m$ is odd, and $n$ is even. Then, the sum $m+n$ is odd.
3. For each $n \in \mathbb{N}$, $\frac{n(n+1)}{2}$ is an integer.
   Suppose for contradiction, that $\frac{n(n+1)}{2}$ is not an integer for some $n \in \mathbb{N}$. Then $n(n+1)$ is odd, but as shown in 1, $n(n+1)$ must be even for each $n \in \mathbb{N}$, which is absurd.

**Example 3.** Prove that two complex numbers $w$ and $z$ are equal if and only if for every $\epsilon > 0$, it follows $|z - w| < \epsilon$. Since this is an "if and only if," in short, iff statement, we must prove both implications, namely necessity and sufficiency.

1. Direct Proof – "only if" part:

   *Proof.* If $z = w$, it follows $|z - w| = 0$. Therefore, for every $\epsilon > 0$, $|z - w| < \epsilon$ holds. ■

2. Proof by Contrapositive – "if" part:
   The "if" part in universally quantified implication form is:

$$\forall \epsilon > 0 : |z - w| < \epsilon \Rightarrow z = w. \tag{5}$$

   The contrapositive form is:

$$z \neq w \Rightarrow \exists \epsilon > 0 : |z - w| \geq \epsilon. \tag{6}$$

   *Proof.* Assume $z \neq w$. Let $\epsilon := |z - w|$. Since $z \neq w$, $\epsilon > 0$. Thus, $|z - w| \geq \epsilon$. ■

3. Proof by Contradiction – "if" part:

   *Proof.* Suppose $|z - w| < \epsilon$ for every $\epsilon > 0$, but assume $z \neq w$ for the sake of contradiction. Let $\epsilon_0 := |z - w|$. Since $z \neq w$, $\epsilon_0 > 0$. By definition $|z - w| = \epsilon_0$, but by hypothesis, $|z - w| < \epsilon_0$ , which is absurd. ■

**Quick Self-Check**

- ☐ The claim is clearly stated, and all symbols and abbreviations are explicitly defined.

  ...the initial statement is the best place to begin your thought process [Cen21]

- ☐ The proof begins with "*Proof.*" and ends with a *QED* symbol such as "■".
- ☐ The type of each variable is indicated: universally quantified ($\forall$, "for all") or existentially quantified ($\exists$, "there exists").
- ☐ The proof type is appropriately chosen: direct, contrapositive, or contradiction.
- ☐ Each logical step follows from previous ones, definitions, or known results.

### 3.2.2 Grammatical Rules

Standard English grammar rules apply to mathematical writing.

- ☐ Every sentence begins with a word, not with a mathematical symbol or an equation.
  For example, instead "$I = [0, 1]$ is compact," write "The unit interval $I = [0, 1]$ is compact." Write "If $k^2$ is even, then $k$ is even." instead of "$k^2$ is even implies $k$ is even."
- ☐ All sentences end with appropriate punctuation.
- ☐ Quantifiers $\forall$ and $\exists$ are carefully used in formal writing.
  Words like "for all" and "there exists" are often better than symbols.
- ☐ Take advantage of mathematical notation.
  For example, instead "let $n$ be a member of the set of natural numbers" write "let $n \in \mathbb{N}$."

**Example 4.** For any non-negative integer $n$, namely for $n \in \mathbb{N}$, prove that $n^3 + 2n$ is divisible by 3.

*Proof.* We will prove this claim using mathematical induction:

- Base Case
  If $n = 0$, then $n^3 + 2n\big|_{n=0} = 0$.
- Induction Step
  Let $n \in \mathbb{N}$. Assume $n^3 + 2n$ is divisible by 3. This means there exists some $m \in \mathbb{N}$ such that

  $$n^3 + 2n = 3m. \tag{7}$$

  Consider $(n+1)^3 + 2(n+1)$:

  $$\begin{aligned}(n+1)^3 + 2(n+1) &= n^3 + 3n^2 + 3n + 1 + 2n + 2 \\ &= n^3 + 2n + 3(n^2 + n + 1) \\ &= 3m + 3(n^2 + n + 1).\end{aligned} \tag{8}$$

  Since both $m$ and $n$ are integers, $(n+1)^3 + 2(n+1) = 3\left(m + n^2 + n + 1\right)$ is divisible by 3.

By the principle of mathematical induction, $n^3 + 2n$ is divisible by 3 for all $n \in \mathbb{N}$. ■

# 4 Report Writing – A Student Checklist

Report writing is a key part of physics laboratory courses. This chapter presents a concise, student-owned checklist to verify completeness and clarity before submission.

> ...reports are heavily organized, commonly with tables of contents and copious headings and subheadings. This makes it easier for readers to scan reports for the information they're looking for. [Ell14]

We adapt, from [Kez10], the following "ABC" format:

(A) A Title Page
- Title of the report
- Names
- Date of lab performance

(B) Body
- Learning Objectives
- Main Concepts and Formulae as Theoretical Background
- Procedure
- Experimental Data and Graphical Analysis
- Sample Calculations and Error Analysis

(C) Conclusion and Citations

Here we present a minimal checklist.

**Basic Slogan: Can a classmate who missed the lab understand the physics and the experiment from your report alone?**

- A title page
  - ☐ The title clearly reflects the main focus of the lab activity and homework analysis.
  - ☐ Names of all collaborators are listed.
  - ☐ Date of **lab performance** is included.

  > When you turn in a work bearing your name, the expectation is that you are submitting work that you have done on your own. That's part of what putting your name on an assignment means: that the work contained inside it is yours, unless expressly stated otherwise. [Bai17]

- Body
  - ☐ Scientific goals are stated briefly in your own words.
  - ☐ In the theoretical background section, key concepts, fundamental definitions, and important equations needed to understand the experiment are presented.
    - ☐ The basic equations actually used in analysis are **derived and explained** in your voice.
    - ☐ All symbols are declared and defined.
    - ☐ Reasoning is clear and logically ordered.
    - ☐ When you say "the" slope, both axes are specified.
  - ☐ A **step-by-step** description of what you actually did is presented in the procedure section.
    - ☐ Useful photos/diagrams, with captions, are included if they aid clarity.
  - ☐ Data tables are all **reader-friendly**.
    - ☐ Tables and figures are numbered in order of appearance with descriptive captions.
    - ☐ Titles, labels, and units are aligned, following the standard.
    - ☐ Numerical precision – sig.fig. – is appropriate.
    - ☐ Cross-references use numbers; write "See table 1" instead "above/below."
    - ☐ SI conventions follow NIST guideline; check [TT19].
    - ☐ Caption goes below the figure and caption goes above the table.
  - ☐ At least one complete example for each core calculation type is presented.
    - ☐ Uncertainty analysis and error propagation are included where appropriate.
- Conclusion and Citation
  - ☐ The conclusion section answers the learning objectives.
    - ☐ It is supported by quantitative evidence: numbers, fits, uncertainties.
    - ☐ A statement of discrepancy, if any, is stated.
  - ☐ Any external ideas, figures, or text are credited appropriately.

- ☐ An appropriate style is selected, see [Por17a].

  A "citation" is the way you tell your readers that certain material in your work came from another source. [Por17b]

- **Your Additional Checkpoints:**
  - ☐
  - ☐
  - ☐

# A Sample Email Template

The following is a general-purpose template for a clear and professional email. Replace each parenthesized placeholder, such as `(COURSE ID)` with your own information.

```
------------------------------------------------------------------------
Subject: (COURSE ID) - Question About (ASSIGNMENT NAME), (PROBLEM NUMBER)

Dear Professor (PROFESSOR NAME),

I am (FULL NAME), in your (COURSE ID, DATE&TIME).
I have a question about (ASSIGNMENT NAME), specifically ...
    (SHORT DESCRIPTION OF ISSUE).

Could you please clarify whether we should use (OPTION A) or (OPTION B)?

Thank you very much,
(FULL NAME)
------------------------------------------------------------------------
```

# References


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